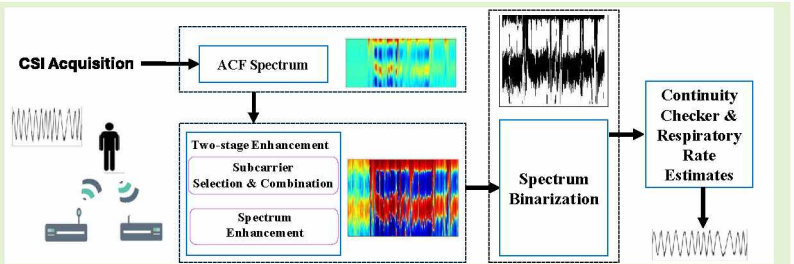


WiResP: A Robust WiFi-Based Respiration Monitoring via Spectrum Enhancement

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Abstract—Respiration monitoring has been attracting substantial attention because of its potential for assessing sleep stages and quality. Traditional approaches for respiratory rate tracking require dedicated wearable devices, which can intrude upon and create an unwelcome experience for users. To address this, researchers have proposed WiFi-based respiration monitoring systems which capitalize on WiFi's ubiquity, cost-effectiveness, and non-contact nature, effectively converting existing infrastructure into ubiquitous sensors. However, existing systems often struggle with the subtle signal to noise ratio of embedded breathing signals, leading to limited coverage and inflexible deployment in noisy environments. This paper presents WiResP (WiFi Respiration Perception), an ingenious and pragmatic WiFi-based system for tracking respiration, employing the spectrum enhancement approach to boost respiration detection. By treating the spectrum of the breathing signal obtained from channel state information as an image, the system is able to leverage image processing techniques to greatly enhance the respiration signal trace, improving detectability and increasing sensing coverage for respiratory rate estimation. Moreover, an image-based continuity checker module is proposed to verify the signal trace continuity to reduce false alarms. We conducted extensive experiments and assessed WiResP under different settings. The experiments demonstrate that WiResP can reliably capture respiratory rate during sleep and achieve a detection rate of 92% and a false alarm rate $< 5\%$, leading to significantly improved sleep stage recognition under flexible device placements. The promising performance positions WiResP as a candidate for real-world in-home respiration tracking system.



Index Terms—Channel State Information, Internet of Things, Respiration monitoring, Spectrum enhancement, Sensors, WiFi sensing.

I. INTRODUCTION

RESPIRATORY rate (RR) is a crucial indicator for a person's health status, for example, the quality and depth of sleep [1] can be determined through sleep stage [2] analysis with RR. IoT-based respiration monitoring have been spurred with the advancements in sensing, communication technology, and low-cost microcomputers. Nevertheless, most existing systems [3]–[5] for RR monitoring, such as chest belts, electrocardiogram devices, and nasal masks, are invasive and uncomfortable due to physical contact, leading to compromised accuracy. Several contactless (device-free) RR monitoring systems have, therefore, been proposed by using dedicate camera [6], [7], ultra-wideband impulse radio [8], [9],

continuous-wave radar [10], [11], millimeter wave radar [12], [13], etc. Although these may offer accurate RR estimation, the specialized devices required can be expensive, rendering commercial deployment infeasible.

Over the past decade, WiFi sensing [14]–[19] has been attracting an increasing attention because of its non-intrusive, privacy-preserving, and cost-effective nature, achieved by repurposing existing infrastructure as ubiquitous sensors. These position it as an appealing option for a wide range of applications, including indoor localization [20], [21], motion detection and classification [22]–[24], and vital signs and sleep monitoring [25]–[43]. Among these, WiFi-based RR estimation has emerged as a particularly popular topic, with numerous studies and proposals dedicated to its development.

WiFi-based RR estimation can be primarily classified into two categories: Received Signal-Strength-Indicator-based (RSSI-based) and Channel State Information-based (CSI-based) methods. RSSI-based RR estimation relies on variations in WiFi signals' received strength caused by chest movements during breathing. This method tracks changes in signal strength over time to calculate RR [35]–[38], [44]; on the other hand, CSI-based RR estimation uses information related to the phase and amplitude of the WiFi signal because CSI

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provides finer and richer information than RSSI, making it capable of capturing more intricate breathing patterns and can be harnessed for various applications beyond RR estimation including analysis of sleep patterns and stages, etc [25]–[34], [39]–[43].

While previous studies [25]–[34], [39]–[43] have demonstrated the potential of RR estimation using WiFi, many of these systems exhibit limited effectiveness in real-world scenarios, relying on controlled settings and specific device deployment configurations. These limitations become pronounced when the strength of the breathing signal is weak, resulting in a low Signal to Noise Ratio (SNR). For example, some of these techniques may demand a direct Line-of-Sight (LOS) connection between the monitored individual and the WiFi transceiver. Furthermore, their operational range is often restricted, necessitating the subject's close proximity to the WiFi devices or links, which stems from the fact that the subtle chest movements are associated with breathing and may not be sufficiently robust for detection unless the devices are optimally positioned. These methods may even falter if the breathing signal is extremely weak when the individual is too far away from the devices or the system is operating in a Non-Line-of-Sight (NLOS) scenario. Such limitations hinder deployment flexibility and restrict the scope of sensing coverage.

In this paper, we propose WiResP (**WiFi Respiration Perception**), a WiFi-based respiration monitoring system, designed to overcome the challenges of inflexible device placement and limited sensing range. WiResP is a novel, training-free and practical image-inspired RR tracking system that can reliably capture the breathing signal, even under low SNR conditions. WiResP estimates RR based on the Auto Correlation Function (ACF) spectrum that has been shown to reflect breathing rates in an environment-independent way [22] and further treats the spectrum as an image and applies a spectrum enhancement technique to highlight/sharpen the trace of the respiration signal. WiResP integrates instantaneous and time-domain data to enhance the recognition of peaks and identification of breathing patterns, markedly enhancing the ability to detect and expanding sensing range for RR estimations. Besides, we designed a false alarm (FA) removal module utilizing breath pattern on ACF spectrum (the continuity of the breath trace) to mitigate false alarms. The extensive experiment evaluation has demonstrated the superior improvement of WiResP on the sensing coverage and deployment flexibility, providing reliable and precise respiration monitoring for sleep stage recognition in real-world settings through WiFi sensing.

The primary contributions of this paper encompass:

- We introduce WiResP, which provides wide sensing coverage for an entire house and allows flexible device placement.
- We use spectrum enhancement techniques to amplify subtle breathing patterns, and consider both instant and time-domain information by leveraging the continuity of breath patterns on the ACF spectrum.
- We assessed WiResP's performance through extensive experiments in various deployment scenarios through detection rate and sleep stage analysis. WiResP surpassed

two reference systems with a 25% higher detection rate and a false alarm rate below 5%.

- To the best of our knowledge, the proposed system is the first to enhance the RR tracking/estimation through image processing techniques by modeling the ACF spectrum as images.

The remainder of this paper is as follows. Section II reviews the related works on WiFi-based RR estimation. Section III provides the fundamental theory of CSI model. Section IV provides an overview of WiResP. Section V discusses the details of WiResP. Section VI shows the system evaluation and experiments results. We delve into constraints of the proposed approach and explore some potential directions for future development in Section VII. Section VIII concludes the paper.

II. RELATED WORK

In this section, we review the related works on contactless RR monitoring.

A. Contactless RR monitoring

Early contactless RR estimation methods include those adopting dedicated camera [6], [7], ultra-wideband impulse radio [8], [9], continuous-wave radar [10], [11], and millimeter wave radar [12], [13]. Vision-based systems using cameras estimate RR by analyzing the difference between frames to track chest and abdomen movements. However, they raise privacy concerns and are sensitive to lighting conditions, necessitating specific placement and orientation. On the other hand, radar-based methods transmit pulses to the target and analyze reflected signals to estimate periodic changes due to breathing motions. Although they provide accurate estimation and are even able to capture the heart beat [12], [13], they typically involve specialized devices, which limits their practicality for home environments. Additionally, radar signals rapidly decay over distance and are directional, requiring precise device orientation and alignment with the area of interest.

B. WiFi-based RR monitoring

The wide deployment of WiFi has sparked significant research on WiFi sensing, giving rise to numerous WiFi-based RR monitoring applications. The authors of [35]–[38] estimate RR using RSSI by analyzing their changes due to breathing; however, the low resolution of RSSI leads to poor accuracy. Wi-Sleep [31] leveraged the amplitude of CSI to estimate respiration, as CSI provides more fine-grained information compared to RSSI. Phasebeat [25] and TensorBeat [34] utilized the phase difference between two antennas to estimate RR. Later, Fresnel model is introduced [40] to explain the respiration detectability with respect to individual's location, orientation and breathing depths for indoor WiFi propagation. TR-BREATH [33] proposes to utilize time-reversal resonating strength (TRRS) and root-MUSIC to estimate RR. Several learning-based techniques [41]–[43] have also been proposed for accurately estimating breathing patterns from noisy radio signals. In addition, various works have investigated multi-user

RR and heartbeats estimation using WiFi [27]–[30], [33], [34], [45].

Many of these systems rely on specific device arrangements and controlled settings, often failing to accurately capture RR when the breathing signal is weak (low SNR). They may require LOS between the tester and the transceiver and have a limited operational range, demanding the subject's close proximity to the device because of the subtle nature of chest movements during breathing. Learning-based methods [41]–[43] often have high accuracy under specific settings, but may require further training in new environments, limiting their applicability. As a result, these systems are often constrained in their coverage area due to their rigid deployment requirements. To extract weak breathing signals and improve the sensing capability, SMARS [46] employs Maximal Ratio Combining (MRC) to combine signals from multiple subcarriers and enhances the SNR of the respiratory signal, whereas FarSense [39] leverages the ratio of the CSIs between two antennas. However, experimental results indicate that the effectiveness of these systems remains susceptible when human subjects are far away from the WiFi devices, constraining their coverage.

In our previous work [47], we have demonstrated the efficacy of spectrum enhancement in enhancing detection rates for room-level scenarios, where devices and the tester are in the same room. However, in house-level scenarios where devices are placed outside the room, we encountered an increased susceptibility to ambient motions and noise, leading to a noticeable increase in false alarms (FAs) due to the broader coverage area when spectrum enhancement was applied. These FAs cannot be removed by the variance filter in our previous work. To address this challenge, this paper expands our prior work by introducing a more comprehensive and robust system. The proposed system further incorporates motion detection as well as a dedicated module to mitigate FAs by considering the continuity of the ACF spectrum from image-processing perspective. Extensive experiments have been conducted to justify its performance.

III. CSI MODEL

Given a pair of a transmitter (TX) and a receiver (RX), the CSI at a certain frequency f on link m at time instance t is modelled [46] as

$$H^{(m)}(f, t) = \sum_{i=1}^L a_i(t) \exp\left(-j2\pi f \frac{d_i(t)}{c}\right) + n(f, t), \quad (1)$$

where $d_i(t)$ is the propagation distance of the i^{th} path and $a_i(t)$ is the amplitude, c denotes the light speed while L denotes the overall count of multipaths. Additionally, $n(f, t)$ is the additive white noise.

As illustrated in Fig. 1, wireless signals traverse via different multipaths, each with a unique sensitivity to environmental changes. The degree of sensitivity is location-dependent, meaning that the presence of motion sources can cause some multipaths to experience time-varying propagation paths and gains, while other paths remain static, allowing the CSI to be

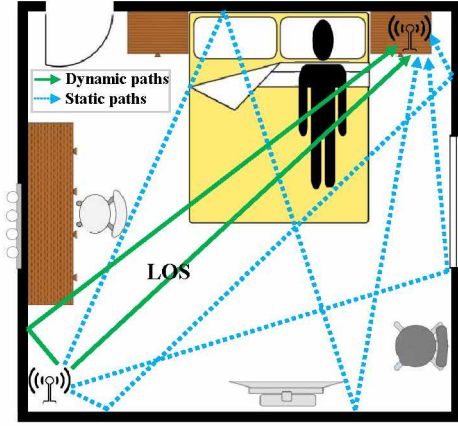


Fig. 1: Indoor Multipath Propagation.

expressed as

$$\begin{aligned} H^{(m)}(f, t) &= \underbrace{\sum_{i \in \Omega_S} a_i e^{-j2\pi f \frac{d_i}{c}}}_{\text{static paths}} \\ &+ \underbrace{\sum_{i \in \Omega_D(t)} a_i(t) e^{-j2\pi f \frac{d_i(t)}{c}}}_{\text{dynamic paths}} + n(f, t) \quad (2) \\ &= H_S^{(m)}(f) + H_D^{(m)}(f, t) + n(f, t), \end{aligned}$$

where $\Omega_D(t)$ denotes the collection of dynamic multipaths affected by time-varying scatters while Ω_S denotes the collection of stationary paths. During a sufficiently short time frame, [46] shows that the stationary paths remain relatively consistent and can be eliminated through mean subtraction, i.e.,

$$\hat{H}^{(m)}(f, t) = H_D^{(m)}(f, t) + n(f, t), \quad (3)$$

where $\hat{H}^{(m)}(t, f_k)$ represents the CSI on link m at t and f with a mean value of zero. For multi-carrier WiFi systems with N_s subcarriers on each link at t , the CSI can be expressed as

$$\hat{H}^{(m)}(f_k, t) = H_D^{(m)}(f_k, t) + n(f_k, t), \quad k = 1, \dots, N_s. \quad (4)$$

$\hat{H}^{(m)}(f_k, t)$ exhibits periodic changes in the presence of periodic environmental fluctuations, such as breathing movements, enabling the RR estimation.

IV. SYSTEM OVERVIEW

This section provides an overview of WiResP illustrated in Fig. 2. WiResP is a WiFi-based sensing system leveraging CSI to detect and estimate RR using only one TX-RX pair. The main components include CSI pre-processing, a motion detector, and a respiration monitoring module.

The CSI pre-processing involves the sanitization of CSI, followed by the calculation of ACF for the subsequent motion detection and respiration monitoring module.

The motion detector is designed to check the surrounding environment for any significant movements. If no movements are detected, the extracted ACF is passed to the respiration monitoring module. The reason is that the estimation of RR

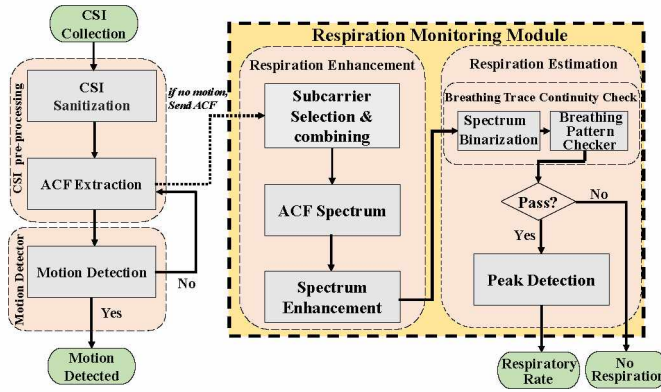


Fig. 2: System Overview

is often compromised due to the disruption generated by movement. This interference disrupts the accurate measurement of subtle changes in WiFi signals, leading to inaccuracies/undetectability in RR estimation.

The respiration monitoring module comprises of two primary elements: (i) respiration enhancement module and (ii) respiration estimation module. In the respiration enhancement module, subcarrier selection and combining techniques are first employed to boost the breathing pattern. Secondly, spectrum enhancement is performed to the ACF spectrum, which is generated by concatenating snapshots of ACF within a given time window.

After that, WiResP performs a continuity check to eliminate FAs and ensure the estimated RR is accurate and reliable, including binarizing the enhanced ACF spectrum to simplify the analysis, and a breathing pattern checker to validate the breathing pattern. If the breathing pattern passes the continuity check, the ACF spectrum is sent to the peak detection module to estimate RR.

V. SYSTEM DESIGN

The system design of WiResP is presented in this section. Assume that the RX has N_r antennas and the TX has N_t antennas, resulting in a total of $N_t \times N_r$ links. Each link comprises N_s subcarriers. The sounding rate is set at F_s Hz and the respiration extraction employs a window size T_w .

A. CSI Pre-processing

CSI measurements are vulnerable to corruption by noise, hardware imperfections, etc. Additionally, CSI values may fluctuate because of the receiver's automatic gain control mechanism. Hence, pre-processing is essential to extract valuable insights. We normalize the CSI samples across all subcarriers and a median filter is applied to each normalized subcarrier over a time window T_w to remove outliers. Next, we extract only the amplitude of the CSI measurements for further processing as the phase of WiFi signals is often too noisy to be useful due to carrier frequency offset and measurement noise. As a result, there are $N_t N_r N_s$ amplitude streams for extracting breathing patterns. Afterwards, the ACF is then calculated for each stream for motion detection and RR estimation later. By computing the ACF of CSI, we can identify the lag at which the signal correlates with itself. The lag at which the ACF

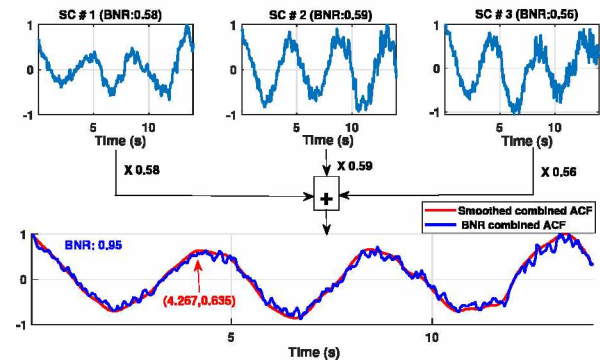


Fig. 3: Depiction of subcarrier combining. The first peak is at $\hat{\tau} = 4.267$, and the estimated RR is $\frac{60}{4.267} = 14.06$ BPM.

exhibits a significant peak corresponds to the period of the signal; thus, RR can be estimated. The choice to utilize ACF was driven by its capability to produce immediate estimations within a short window [46] and it has been shown to reflect breathing rates in an environment-independent way [22]. To be specific, denote $s_i^{(m)}$ as the CSI vector of the i^{th} subcarrier on the link (n_t, n_r) , the ACF from amplitude, $\rho_i^{(n_t, n_r)}(\tau)$ is computed as

$$\rho_i^{(n_t, n_r)}(\tau) = \frac{COV[|s_i^{(n_t, n_r)}(t)|, |s_i^{(n_t, n_r)}(t + \tau)|]}{COV[|s_i^{(n_t, n_r)}(t)|, |s_i^{(n_t, n_r)}(t)|]}, \quad (5)$$

where $\tau = 0, \dots, T_w F_s$ represents the time lag and COV stands for the operation of covariance. In the sequel, $\rho_i^{(m)}(\tau)$ is used to represent the ACF obtained from amplitude streams and $m = 1, \dots, N_t N_r N_s$.

B. Motion Detector

Because breathing signals can be easily corrupted by large motions in the environment, leading to inaccurate respiration rate detection, respiration extraction should not be performed during the presence of significant motions. Therefore, prior to RR estimation, we can adopt an ACF-based motion detection as proposed in our earlier work [22] for lower system complexity and power consumption. As shown in [22], the first tap of the ACF obtained from CSI power, known as motion statistics ξ , indicates the overall strength of the surrounding motions, and in practice can be approximated by evaluating at $\tau = 1/F_s$ as below

$$\xi = \mathbf{E} \left[\rho_i^{(m)}(\tau) \right] \Big|_{\tau = \frac{1}{F_s}}, \quad (6)$$

where $\mathbf{E}$ is the expectation operator and $m = 1 \dots N_t N_r N_s$. Then, the respiration tracking is only enabled if ξ is less than a predetermined threshold η , which was empirically chosen by evaluating typical daily human motions, excluding static activities [22].

C. Respiration Signal Enhancement

The section presents an enhancement module with a two-phase configuration for boosting SNR of the respiration signal.

1) **Phase 1 - Subcarrier selection and combining:** Subcarrier selection and combining can be challenging because of the fluctuating impacts of multipath on individual subcarriers. Merely computing the average across all subcarriers may not be effective in improving the combined signal's SNR, particularly when the tester is far from the link. In such cases, certain subcarriers might experience greater corruption than others, as shown in [46] [48]. To overcome this, we propose a solution that involves selecting¹ K subcarriers out of the total $N_s N_t N_r$ based on their highest Breathing-to-Noise Ratio (BNR) [49]. The BNR is the proportion of energy within and beyond the range of interest, specifically pertaining to normal human respiration². The combined ACF can be obtained as $\rho_{combined} = \sum_{i \in \mathcal{V}} \rho_i \times BNR_i$, where $\mathcal{V}$ refers to the collection comprising the foremost K candidates exhibiting the highest BNR. Next, we apply local estimated scatterplot smoothing (LOESS) to refine the merged ACF signal [46]. An instance of a merged and smoothed ACF is shown in Fig. 3, demonstrating a significant improvement in BNR.

2) **Phase 2 - ACF spectrum enhancement:** An ACF spectrum is obtained by concatenating ACFs over consecutive time windows. When valid breathing occurs over this time span, the concatenated ACFs form discernible breathing traces with interconnected peaks and valleys representing the periodicity of the breathing signal. Fig. 4(a) displays an ACF spectrum extracted from 20-minute of breathing data, showcasing distinct respiratory patterns in certain intervals. Nevertheless, during certain intervals, the breathing signal is not strong enough to produce a detectable pattern, leading to an obscured ACF spectrum and undetected RRs due to various reasons, such as the subject changing their sleeping posture or experiencing shallow breathing during a particular sleep stage.

To improve peak recognition and breath pattern identification, we introduce an approach that considers the ACF spectrum in the form of an image. We apply the histogram equalization method to enhance the SNR of breathing signals. This technique distributes the intensities uniformly and increases contrast, enabling better breath pattern identification. Let T_e be the enhancement window size, measured in seconds and $X = [x_1, \dots, x_{T_w F_s \times T_e F_s}]$ represent the array of ACF spectrum. The calculated value of the transformed variable y_i is

$$\hat{F}_X(t) = \frac{\sum_{i=1}^{F_s^2 T_e T_w} \mathbb{1}_{(x_i \leq t)}}{F_s T_w \cdot F_s T_e},$$

$$y_i = \hat{F}_X(x_i), \quad (7)$$

where $\hat{F}_X(\cdot)$ represents the empirical cumulative distribution function (CDF) of X , and $\mathbb{1}(\cdot)$ denotes the indicator function. Fig. 4(b) shows that the light and weak breathing signal trace becomes more distinguishable after applying histogram equalization.

¹ K can be chosen through empirical methods, with K usually being much smaller than $N_s N_t N_r$.

²The standard breathing rate for an average human ranges from 10 to 30 breaths per minute (BPM).

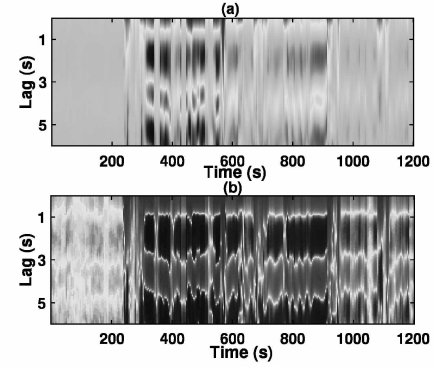


Fig. 4: (a) Original spectrum. (b) Enhanced spectrum.

D. Breathing Trace Continuity Examination

Although the above techniques can significantly improve RR detection capability, it is still susceptible to false alarms caused by ambient small motions ($\xi < \eta$) or noise amplified by the subcarrier combining and histogram equalizer, which can create a breathing-like ACF and thus lead to false detection. Therefore, it is important to only estimate the RR when an actual RR is detected which sustains for a certain period.

We propose a low-complexity pattern-based breathing trace continuity check which employs Otsu's threshold method [50] to binarize the ACF spectrum obtained in Section V-C.2. Spectrum binarization simplifies the identification of peaks and valleys in the enhanced breathing pattern, enabling easier comparison with the reference signal, therefore, improving the analysis of the enhanced ACF spectrum and reducing computational complexity.

The resulting binarized ACF spectrum is then sent to the breathing pattern checker, which averages the binarized spectrum over a duration of T_e to obtain the overall trend. The overall trend is then compared with a reference binarized 1-D breathing pattern. The following shows the steps.

• Step1:

Divide the enhanced ACF spectrum $Y = [y_1, \dots, y_i, \dots, y_{T_w F_s \times T_e F_s}]$ in (7) in two classes C_0 and C_1 with the threshold Θ determined by the below,

$$\Theta = \arg \max_{0 < \theta < 1} \hat{F}_Y(\theta)[1 - \hat{F}_Y(\theta)] (\mu_{C_0} - \mu_{C_1})^2, \quad (8)$$

where μ_{C_0} and μ_{C_1} denote the mean of C_0 and C_1 respectively; $\hat{F}_Y(\cdot)$ is the empirical CDF of the enhanced spectrum Y . Then the binarized spectrum $\mathcal{Y} = \{\mathcal{Y}_i\}$ can be expressed as

$$\mathcal{Y}_i = \begin{cases} 1, & y_i \in C_0 = \{a_i | a_i \geq \Theta\} \\ 0, & y_i \in C_1 = \{a_i | a_i < \Theta\} \end{cases}. \quad (9)$$

Fig. 5(a) shows an example of breathing ACF spectrum, and Fig. 5(c) shows the binarized spectrum of Fig. 5(a).

• Step2:

The binarized ACF spectrum is averaged over a

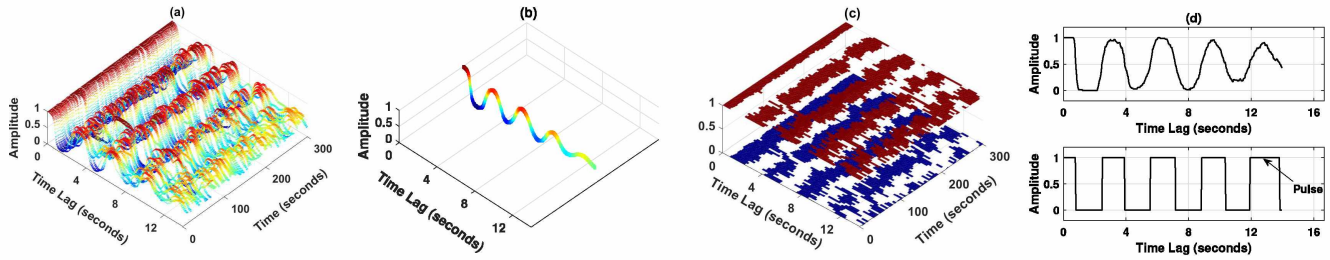


Fig. 5: (a) Breathing ACF spectrum over time. (b) The overall average of (a). (c) The binarized ACF spectrum of (a). (d) The trend obtained from (c) and its binarized waveform.

duration of T_e to assess the continuity of breathing traces³. To be specific, denote $\tilde{\mathbf{u}}_t$ as the snapshot of binarized ACF at time t . The averaged binarized ACF spectrum is $\frac{1}{T_e} \sum_{t=1}^{T_e} \tilde{\mathbf{u}}_t$. The top figure of Fig. 5(d) shows the averaged binarized ACF spectrum.

• **Step3:**

Binarizing again the averaged binary ACF obtained in step 2 to reduce the impact of short-term fluctuations provides an 1-D averaged binary ACF, $\tilde{\mathbf{u}}_B$ as Fig. 5(d) shows. To be specific, the 1-D signal is calculated via

$$\tilde{\mathbf{u}}_B = \mathcal{B} \left(\frac{1}{T_e} \sum_{t=1}^{T_e} \tilde{\mathbf{u}}_t \right),$$

$$\mathcal{B}(v) = \delta_{v \geq \frac{1}{2}}, \quad (10)$$

where $\mathcal{B}(\cdot)$ is element-wise operator that rounds each element of the input and δ is Kronecker delta function.

This trend $\tilde{\mathbf{u}}_B$, is then utilized to ascertain the presence of valid breathing within the time frame of T_e , based on the continuity of a normal respiratory pattern. If a consistent respiratory pattern is detected throughout T_e , it is deemed a valid breathing event, from which the RR can be estimated. For instance, the number of breathing cycles, i.e., the rectangular pulses depicted in Fig. 5(d) and the average pulse width within T_w can serve as robust indicators of normal human respiration. It is important to note that the original ACF spectrum, i.e., Fig. 5(b) should not be directly averaged as the amplitude of the ACF fluctuates over time, making it difficult to compare with a reference signal. With the binarized waveform, comparison with a reference signal is simplified.

E. Respiratory Rate Estimation

After valid breathing is detected within the timeframe T_e , the RR is estimated using the enhanced spectrum through the time difference between the first peak and 0. For example, Fig. 3 illustrates an instance of τ . We use the enhanced spectrum instead of the binarized spectrum to estimate the RR because the binarized spectrum has information loss compared to the original enhanced spectrum with clear peak/valley, particularly in applications that necessitate precise RR estimation, such as sleep stage analysis.

³ T_e typically contains 5 ~ 7 breathing cycles.

VI. EXPERIMENTS RESULTS

This section presents the evaluation of WiResP with commercial WiFi chipsets. Extensive experiments were conducted in home environments.

A. Experiment Setup

We evaluated WiResP within a real-world residential environment with a set of commercial WiFi devices employing a Qualcomm WiFi module that supports 802.11ac and dual-band functionality on 2.4G and 5G. The devices were set to operate at 5.805 GHz and $F_s = 30$ Hz, with RX/TX having two/one antennas. The bandwidth is 40 MHz, divided into 128 subcarriers with 24 subcarriers reserved for guard bands. We collected 35 overnight sleep data (approximately 280 hours), and used SleepBreathe [51], a home-use sleep monitoring mask, to gather accurate RR as the ground truth.

To demonstrate the system's enhanced coverage and flexible device placement regardless of the condition when the devices/subject are at a fixed location, we carried out comprehensive experiments with varying room dimensions considering various device orientations/locations, and transceiver link distances relative to the tester under natural sleep conditions, including different sleeping positions and whether the person was covered by a quilt. Additionally, we consider both LOS and NLOS with the TX placed both inside and outside the room along with different device heights. We placed two sets of devices in two different rooms with multiple RXs to enhance the data collection efficiency, represented by green circles and brown rectangles in Fig. 6. For each set, we considered three different cases to evaluate performance. We used TX_i^k to denote the TX of set i for case k and $RX_{i,j}^k$ to denote the j^{th} receiver (RX) of set i for case k .

- **Set 1 (green circles):** The positioning of the receivers was guided by the availability of power outlets in testers' rooms and the heights are determined by the heights of power outlets, desks or nightstands, which contrasts with many other studies that necessitate particular device arrangements.

- Case 1 ($RX_{1,j}^1$): A LOS condition with the TX positioned on the nightstand next to the bed. The $RX_{1,6}^1$ is placed to provide a link crossing the body at a close proximity, providing an ideal placement for breathing detection, serving as the baseline.

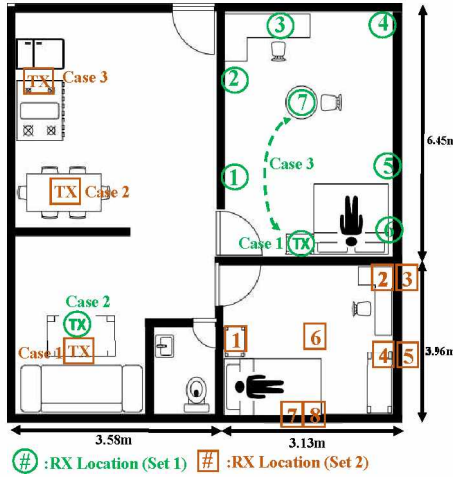


Fig. 6: The illustration of all experiment cases.

- Case 2 ($RX_{1,j}^2$): A NLOS scenario in which the system was evaluated with the TX positioned outside the room where the RXs of Set 1 were placed.
- Case 3 ($RX_{1,j}^3$): This is a LOS condition in which we interchange the positions of RX#7 and the TX in Case 1. We assessed the effects of different link orientations on the system performance.
- **Set 2 (brown rectangles):** we studied RX heights and different locations. We placed RX#1, RX#3, RX#5, and RX#7 at a typical desk height while we mounted RX#2, RX#4, RX#6, and RX#8 at the ceiling.
 - Case 1 ($RX_{2,j}^1$): A NLOS scenario where the TX is near the room where the RXs are deployed. The separation between RX#1 and TX amounts to approximately 2.5 meters.
 - Case 2 ($RX_{2,j}^2$): A NLOS scenario where the TX is positioned on the dining table. The separation between RX#1 and TX amounts to approximately 5 meters.
 - Case 3 ($RX_{2,j}^3$): A NLOS scenario where the TX is positioned on the countertop. The separation between RX#1 and TX amounts to approximately 8 meters.

Note that ($RX_{2,2}^k, RX_{2,3}^k$), ($RX_{2,4}^k, RX_{2,5}^k$), ($RX_{2,7}^k, RX_{2,8}^k$) are located at the same position, but differ in their height placement. Table I summarizes the heights of all the devices, and Fig. 6 illustrates all the cases.

B. Comparison of Detection Rates

1) **Overall Comparison:** Fig. 7 illustrates the overall detection rate with respect to the BNR of the original breathing signal without enhancement across three systems. WiResP attains a detection rate of 90% when the BNR is approximately 3 while FarSense and SMARS reach the same detection rate

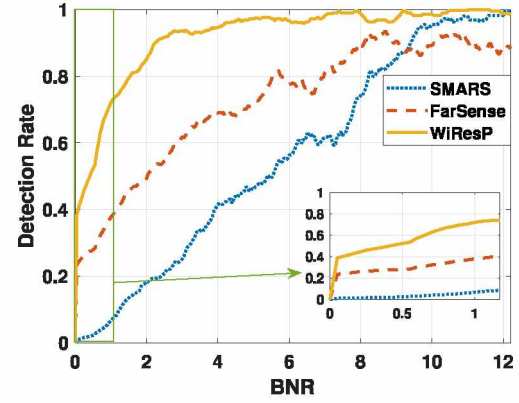


Fig. 7: Detection Rate Versus BNR.

when the BNR is around 8 and 10. The necessary BNR to achieve a 90% detection rate for WiResP has been reduced by approximately 2.5 times compared to the other references. WiResP notably improves the breathing signal traces even at lower BNR. It is worth noting that the detection rates of WiResP can continue to sustain certain performance until the BNR approaches zero, overcoming the major drawbacks of the existing schemes. This is because that, at extremely low BNRs, the quality of the breathing pattern is too poor to be reliably enhanced. The superior performance of WiResP can be attributed to its effective spectrum enhancement technique, which significantly enhances the SNR of the breathing signals.

2) **Detailed Comparison Over Locations:** Fig. 8 shows the detailed comparison on the detection rates of three systems for set 1. WiResP outperforms SMARS and FarSense by over 20% across all locations due to its two-phase SNR boosting technique. It can be seen that RX#6 in case 1 for set 1 ($RX_{1,6}^1$) demonstrates superior performance as the link between the TX and RX passes over the chest area, enabling robust capture of chest motions for RR detection. The detection rate is generally good for room level settings, i.e., case 1 and case 3, as LOS holds and the systems can detect chest motions effectively. However, with an increasing separation between the RX, TX, and the tester, i.e., case 2, the effectiveness of all three systems declines. The detection rate of $RX_{1,1}^1$, $RX_{1,2}^1$, and $RX_{1,5}^1$ is not as high as the other RX locations since the link length between RX and TX is not long enough to consistently cover the tester. Some of these limitations can be partially attributed to the Fresnel zones as addressed in [40]. Fresnel zones consist of concentric ellipsoids with TX and RX as foci. The interaction of multipaths within these zones causes variations in signal amplitude at RX: reflections from odd zones enhance the amplitude, making the response on the CSI waveform more pronounced, while even zones tend to diminish it. Nevertheless, WiResP still outperforms the other

	TX_i^1	TX_i^2	TX_i^3		$RX_{i,1}^k$	$RX_{i,2}^k$	$RX_{i,3}^k$	$RX_{i,4}^k$	$RX_{i,5}^k$	$RX_{i,6}^k$	$RX_{i,7}^k$	$RX_{i,8}^k$
Set 1 ($i = 1$)	0.47m	0.77m	0.75m	Set 1 ($i = 1$)	0.47m	0.47m	0.74m	1.58m	1.14m	0.81m	0.75m	–
Set 2 ($i = 2$)	0.77m	0.95m	0.95m	Set 2 ($i = 2$)	0.47m	2.10m	0.70m	2.10m	0.75m	2.18m	0.80m	2.10m

TABLE I: The heights of TX's and RX's for set 1 & set 2.

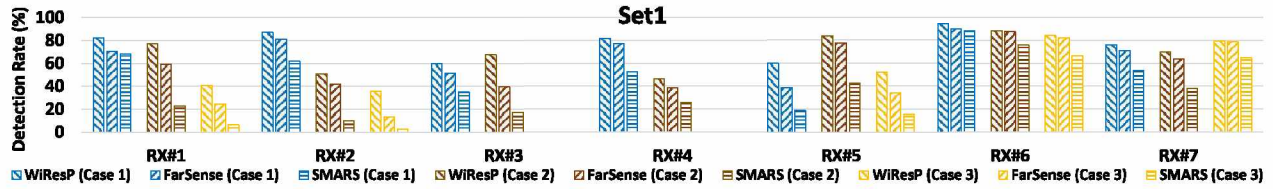
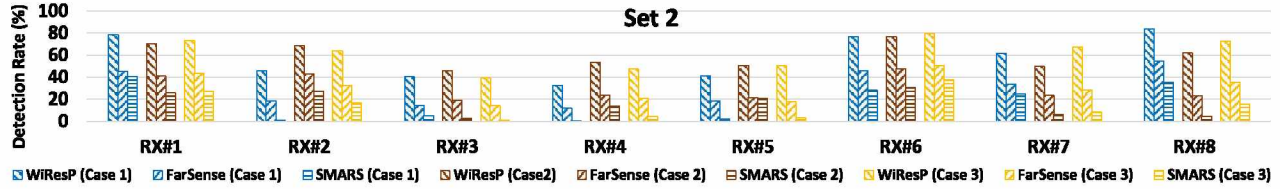
Fig. 8: Detection Rate of set 1 for three cases⁴.

Fig. 9: Detection Rate of set 2 for three cases.

two methods in these challenging scenarios because of its critical enhancement, demonstrating its superior performance.

Fig. 9 evaluates the detection rates under NLOS conditions for set 2. It is evident that WiResP significantly outperforms the other two systems, with a performance improvement of over 25% thanks to the enhancement module. It is worth noting that the detection rate at $RX_{2,3}^k$, $RX_{2,4}^k$, and $RX_{2,5}^k$ for $k = 1 \sim 3$, is generally lower than the other locations because of the increased distance between the transceiver and the tester, resulting in degraded signal strength. This decline in performance can be partly clarified using the concept of the Fresnel Zone, as discussed earlier [40].

C. Sleep Monitoring Analysis

We evaluate the effectiveness of three systems for reliable detection of RR with our previously verified sleep stage analysis through RR [46]. The principle is that the RR generally has higher variability and irregularity during the Rapid Eye Movement (REM) stage while it is lower and more stable during the Non-Rapid Eye Movement (NREM) stage. During the wake stage, body motions are more frequent than during NREM and REM stages but less frequent than during periods of continuous activity, such as non-sleep movements. Therefore, the sleep stage of the tester can be classified into Wake, NREM and REM⁵. We define the active stage as the period with significantly continuous motions from the tester during which the RR cannot be estimated while the inactive stage represents a period where inadequate RR and no motion are detected. This evaluation provides insight into the system stability for respiration monitoring, beyond mere detection rate analysis. Therefore, we employ this sleep analysis through calculating the missed RR detection in sleep monitoring, SM_{missed} as

$$SM_{missed} = \frac{Dur_{\{Inactive\}}}{Dur_{\{Sleep\}} - Dur_{\{Active\}}},$$

⁴Note that in Case 3, the results of RX#3 and RX#4 are excluded because their coverage is too small to reach the tester.

⁵Please refer to the "REM/NREM Recognition" section in [46] for further details.

$$SM_{Pass} = 1 - SM_{missed}, \quad (11)$$

where Dur_{Sleep} is the total duration of each sleep dataset, and $Dur_{\{Active\}}$ and $Dur_{\{Inactive\}}$ are the duration of active stage and inactive stage, respectively. SM_{Pass} denotes the ratio of successful RR detection for sleep monitoring.

Table II showcases the SM_{Pass} for each location in set 1 and set 2. It is clear that WiResP provides a more stable and continuous RR detection and estimation for sleep stage analysis. An interesting point to note is that even though in some scenarios, certain systems show a satisfactory RR detection rate, they still have a relatively inferior SM_{Pass} . This discrepancy can be attributed to the inconsistent and intermittent detection of RR during sleep stage analysis, leading to more frequent classification of inactive stage. For example, while WiResP retains roughly the same detection rate on $RX_{2,1}^1$ and $RX_{2,8}^1$, it demonstrates a lower SM_{Pass} on $RX_{2,1}^1$. Moreover, in certain instances, some systems display a higher SM_{Pass} than their corresponding detection rate. This can be explained by the fact that although RR is not detected every second within a predetermined window for sleep stage analysis, the detections are sufficient for a reliable sleep stage analysis. For instance, WiResP has $SM_{Pass} = 86.27\%$ on $RX_{2,6}^3$ while its corresponding detection rate is 77.61% in Fig. 9. Overall, WiResP can achieve comparable performance of house-level settings to a room-level scenario. This suggests that WiResP is capable of offering extensive sensing coverage while still maintaining stable and accurate RR detection for sleep stage analysis.

To further assess the sleep monitoring performance based on the RR calculated by the three algorithms, Fig. 10 and Fig. 11 show the comparison of sleep stages at a few selected locations for sets 1 ($RX_{1,6}^1$ in green, $RX_{1,2}^1$ in yellow, $RX_{1,3}^1$ in pink) and set 2 ($RX_{2,1}^1$ in green, $RX_{2,8}^1$ in yellow, $RX_{2,6}^1$ in pink) representing the scenarios with different breathing detection capability. $RX_{1,6}^1$ of the WiResP and $RX_{2,1}^1$ of the WiResP are selected as the reference⁶ sleep stages in set 1 and set 2 respectively based on the their robust RR detection rate

⁶The reference locations are in green and the yellow one and the pink one represent the other two selected locations.

	$RX_{1,1}^k$	$RX_{1,2}^k$	$RX_{1,3}^k$	$RX_{1,4}^k$	$RX_{1,5}^k$	$RX_{1,6}^k$	$RX_{1,7}^k$	
WiResP (k=1)	79.75%	84.42%	60.12%	79.31%	57.61%	90.35%	85.38%	
FarSense (k=1)	71.18%	71.78%	50.14%	67.28%	53.93%	89.52%	83.11%	
SMARS (k=1)	72.82%	47.27%	39.66%	49.16%	13.56%	89.61%	71.65%	
	$RX_{2,1}^k$	$RX_{2,2}^k$	$RX_{2,3}^k$	$RX_{2,4}^k$	$RX_{2,5}^k$	$RX_{2,6}^k$	$RX_{2,7}^k$	$RX_{2,8}^k$
WiResP (k=1)	64.6%	48.07%	35.52%	39.8%	30.18%	81.04%	60.07%	85.79%
FarSense (k=1)	29.73%	8.5%	12.67%	5.59%	6.13%	39.63%	22.81%	69.76%
SMARS (k=1)	13.73%	0%	0%	0%	0%	13.24%	15.3%	30.59%
WiResP (k=2)	65.77%	65.79%	51.2%	33.24%	48.22%	75.09%	65.79%	60.28%
FarSense (k=2)	43.75%	36.42%	30.83%	6.07%	30.07%	64.12%	48.60%	3.85%
SMARS (k=2)	36.21%	18.75%	0.00%	0.00%	0.00%	35.82%	25.31%	0.00%
WiResP (k=3)	57.98%	46.59%	—	—	61.95%	80.66%	80.35%	—
FarSense (k=3)	35.36%	21.09%	—	—	21.84%	77.40%	76.04%	—
SMARS (k=3)	0.00%	0.00%	—	—	0.00%	78.45%	62.36%	—

TABLE II: Averaged $SM_{P_{ass}}$ of each location for set 1 & 2.

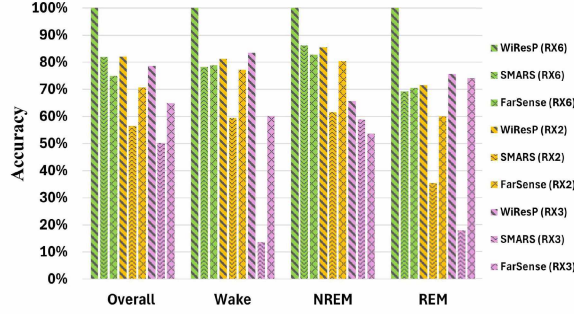


Fig. 10: Comparison of Sleep Stages in Case 1 of Set 1 ($RX6 = RX_{1,6}^1$, $RX2 = RX_{1,2}^1$, $RX3 = RX_{1,3}^1$).

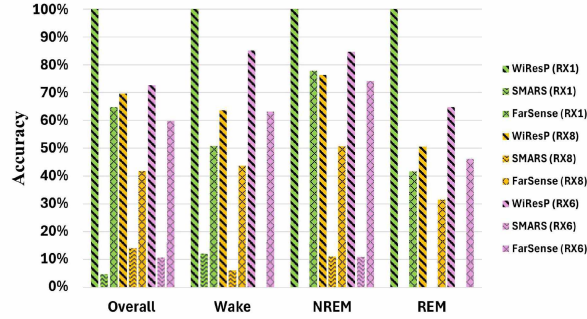


Fig. 11: Comparison of Sleep Stages in Case 1 of Set 2 ($RX1 = RX_{2,1}^1$, $RX8 = RX_{2,8}^1$, $RX6 = RX_{2,6}^1$).

evaluated in VI-B. WiResP demonstrates over 80% overall accuracy relative to the selected reference, $RX_{1,6}^1$ in set 1 even when the device location is positioned away from the tester, i.e., $RX_{1,2}^1$, $RX_{1,3}^1$, while the other methods experience drastic accuracy drop. For more challenging scenarios in set 2 with devices located in separate rooms, WiResP still achieves around 70% relative to the selected reference $RX_{2,1}^1$, outperforming the other two algorithms that show more significant accuracy drops and fail to maintain consistent performance. The ability of WiResP to sustain performance under flexible device placements show its superior capability in achieving continuous and stable RR detection, which is critical for accurate and robust sleep stage analysis, thanks to its larger detection coverage.

D. Performance Gain from BNR-based Subcarrier Selection

In this subsection, we evaluate the effectiveness of the BNR-based subcarrier selection in comparison to the MRC-based

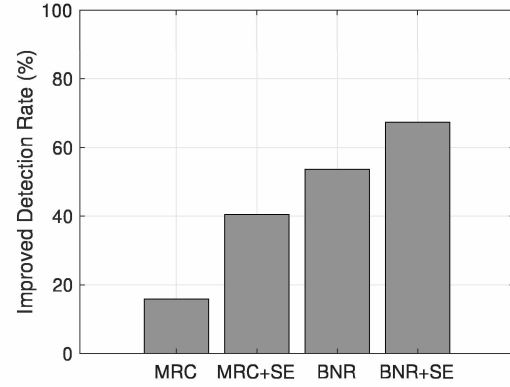


Fig. 12: Performance Gain of different modules.

selection⁷. We measure the average detection rate across all TX and RX locations, and the overall detection rate is shown in Fig. 12, where “MRC+SE” and “MRC” mean the MRC-based methods with and without spectrum enhancement (SE), and “BNR+SE” and “BNR” mean the BNR-based methods with and without SE, respectively. It can be seen that the BNR-based method outperforms the MRC-based approach, resulting in a substantial increase in detection rate. Specifically, BNR outperforms MRC by 38%, and BNR+SE surpasses MRC+SE by 28%. A significant improvement is observed when spectrum enhancement is applied to MRC-based methods. However, the MRC-based subcarrier combination faces challenges in effectively capturing subcarriers containing weak breathing signals, leading to a limited improvement from spectrum enhancement. In contrast, the BNR-based approach successfully identifies and merges these subcarriers to amplify the SNR of the respiratory signal, further improving detectability. Moreover, enhancement can further augment the detection rate when utilizing the BNR-based subcarrier selection and combination method.

E. Performance Gain from Spectrum Enhancement

We evaluate the effectiveness of spectrum enhancement through a comparison of the detection rate with and without it over all RX locations and scenarios. Fig. 12 indicates that spectrum enhancement improved the detection rate significantly. Specifically, MRC+SE outperforms MRC by 26% and BNR+SE outperforms BNR by 14% respectively. Fig.

⁷The averaging method for all subcarriers has been omitted due to its inferior performance, as demonstrated in previous studies cited in [46] and [48].

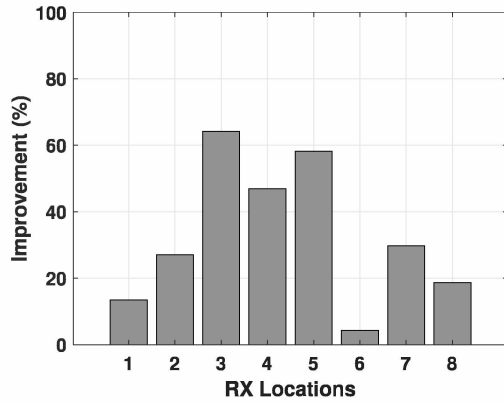


Fig. 13: Improvement at each location in Set 2.

13 shows the detailed improvement achieved by spectrum enhancement for set 2 across different RX locations. The degree of improvement varies depending on the location. RX#6 shows a slight improvement compared to the others due to its central mounting position on the ceiling, which provides more homogeneous coverage of the tester. The improvements from RX#3 to RX#5 are significant. It is worth noting that the subcarrier selection and combining method actually captures the respiration pattern, but the SNR is still too low to detect the pattern, especially when the tester is situated at a considerable distance from the link or devices. WiResP performs second-phase enhancement, spectrum enhancement, to significantly enhance the pattern and SNR, thereby improving detection.

F. False Alarm Mitigation via Trace Continuity Check

Fig. 14 displays the overall FA charts at 8 RX locations in set 2 overall all cases for three methods and compare them with FA rate of WiResP without trace continuity examination. Without the continuity examination module, the FA rate of WiResP is approximately 10% because the enhancement module also enhances the noise, resulting in more FAs. However, the FA rate drops to less than 3% when the trace continuity examination module is incorporated. It is evident that FarSense incurs some false alarms due to the division operator, which amplifies some noise. Conversely, SMARS has the smallest FA because its detectability is weaker compared to the other methods.

G. Precision of Respiratory Rate Estimation

Fig. 15 and Fig. 16 depict the precision of RR estimation for three algorithms at $\text{BNR} \approx 3$ and $\text{BNR} \approx 10$, respectively. A nasal mask [51] was employed for the instantaneous RR reference. When the BNR is low depicted in Fig. 15, both SMARS and FarSense struggle to detect the respiration signal, resulting in poor accuracy. In contrast, WiResP consistently detects and accurately estimates the respiratory rate when the BNR is low, with a median discrepancy of approximately 0.92 BPM and a 90th percentile error of 2.82 BPM. When BNR is high as Fig. 16 shows, three methods are able to detect and estimate RR. The median discrepancy and the error at the 90th percentile of WiResP is only 0.78 BPM and 2.49 BPM.

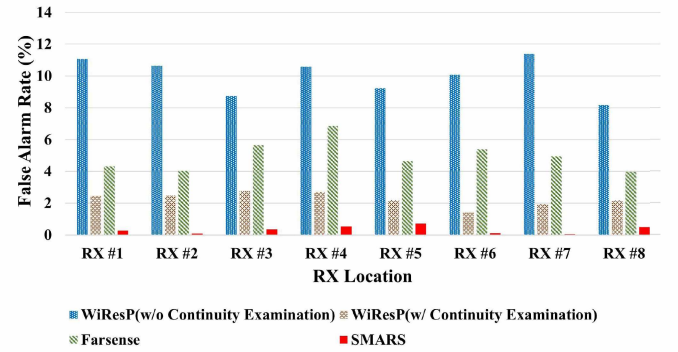


Fig. 14: False alarm rate across three cases in Set 2.

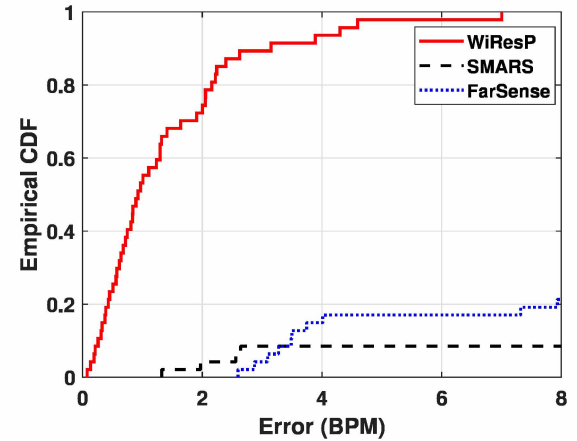


Fig. 15: Precision of RR estimates (BNR ≈ 3).

WiResP outperforms the other two systems in terms of 90th percentile error and median error by approximately 0.1/1.2 BPM in 90th percentile error and 0.1/0.28 BPM in median error.

VII. DISCUSSION AND FUTURE WORK

The section discusses certain constraints of the proposed system as well as potential directions for future research.

A. Complexity Reduction

Complexity reduction is always necessary and important for real-world implementations. Although the proposed system is capable of enlarging the sensing range and allows flexible deployments of the devices, the good performance is naturally at the price of increased complexity. We address the overhead of the modules in the system and give some potential solutions to reduce the complexity.

- **Candidate/Subcarrier Selection:** In WiResP, each subcarrier is Fourier transformed to calculate the BNR for selection, and the computation complexity for one respiration estimate is $\mathcal{O}(N_t N_s N_r N \log N)$ given the FFT size N . The overhead increases with more antennas and larger bandwidth. To reduce complexity, subcarrier selection may not be necessary for each estimate if the subject's location, orientation, posture, and environment

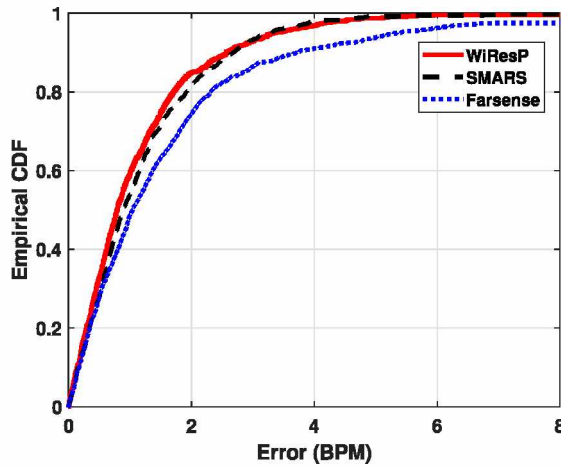


Fig. 16: Precision of RR estimates (BNR ≈ 10).

remain stable. Selection can be based on criteria such as significant motion or environmental changes. Another approach is to select a single, most sensitive antenna combination of TX and RX for respiration estimation.

- **Histogram Equalizer:** The histogram equalization is currently applied even if the original spectrum provides clear respiration patterns, which may lead to unnecessary complexity. A decision-making algorithm can be considered to determine if histogram equalization is needed based on the continuity of the breathing trace and the window size for assessing the original spectrum's quality.

B. Multi-Subjects Respiration Estimation

While our proposed system focuses on respiration detection and estimation for a single person, it can be extended to sense multiple subjects in a room. However, separating multiple respiration signals from radio signals bouncing off in all directions can be an arduous task. Additionally, if the respiration rates of multiple subjects overlap, the task becomes even more challenging. An achievable approach involves the utilization of signal decomposition techniques; for instances, empirical mode decomposition (EMD) [52] and independent component analysis (ICA) [49], which have been used in various fields. We leave this as a future direction for our work.

C. Continuity of ACF Spectrum Recognition

The proposed module for examining the continuity of a breathing trace is shown to greatly reduce FAs through predetermined thresholds, such as the window size T_w , number of pulses/widths, and the reference 1-D breathing pattern. However, it may require further fine-tuning processes depending on specific application requirements and environments. Future research can employ learning-based classifiers with massive ACF spectrum images from diverse cases, including empty and stationary environments as well as environments with sleeping testers exhibiting various breathing patterns, under various sleeping positions, to reduce both false negatives and positives.

VIII. CONCLUSION

We presents WiResP, an innovative respiration monitoring system employing commercial WiFi chipsets for real-world implementations. Through the utilization of spectrum enhancement, WiResP enhances the contrast within the ACF spectrum, augmenting the ability to sense and estimate respiration rates. WiResP surpasses cutting-edge systems in terms of detection rate and accuracy, facilitating respiratory rate monitoring within households without requiring additional hardware or deployment requirements. Our approach represents a novel image processing perspective to tackle the challenge of respiratory tracking and holds promise for extension into other realms of WiFi-based sensing, improving the detection rate more than 25% while the false alarm rate is below 5%. We believe that WiResP vastly simplifies the setup of devices and broadens the reach of sensing coverage, bridging the gap between laboratory and practical deployments of devices. The future work includes testing WiResP in diverse home environments, complexity reduction with intelligent selection of subsets of antennas or subcarriers, and incorporating learning-based methods for further FA mitigation.

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